Solve with us 3.1 - Not Graded

This assignment will not be graded and is only for practice.
Key Points: 1Key Points: 1

Definition of a vector space V: Definition of a vector space V:

- A vector space VV over RR is a set along with two functions:

1. +: $V \times V \rightarrow V$+ : V × V → V (VV is closed under vector addition i.e., if $x, y \in V \implies x + y \in V, \forall\, x, y \in V$x, y ∈ V ⟹ x + y ∈ V , ∀ x, y ∈ V)
2. .: $R \times V \rightarrow V$. : R × V → V (VV is closed under scalar multiplication i.e., if $x \in V$ and $c \in R \implies c.x \in V, \forall\, x \in V,$ and $\forall c \in R$ x ∈ V and c ∈ R ⟹ c.x ∈ V , ∀ x ∈ V , and ∀c ∈ R)

- Axioms of vector additions of a vector space

1. $v_1 + v_2 = v_2 + v_1, \forall\, v_1, v_2 \in V$v1 + v2 = v2 + v1, ∀ v1, v2 ∈ V .
2. $v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3, \forall\, v_1, v_2$ and $v_3 \in V$v1 + (v2 + v3) = (v1 + v2) + v3, ∀ v1, v2 and v3 ∈ V .
3. There exists an element 0 (called the zero vector of VV) in VV such that $0 + v = v, \forall\, v \in V$ 0 + v = v, ∀ v ∈ V .
4. There exists a vector v' v′ in VV such that $v' + v = v + v' = 0, \forall\, v \in V$v′ + v = v + v′ = 0, ∀ v ∈ V (v' v′ is called the inverse (or negative) of vv with respect to addition).

- Axioms relating vector addition and scalar multiplication of a vector space (since this is usual multiplication in real numbers, we suppress the "." sign).

1. For each vector $v \in V$v ∈ V , $1v = v$.1v = v.
2. For each vector $v \in V$v ∈ V and for each pair $a, b \in R, (a + b)v = av + bv$.a, b ∈ R, (a + b)v = av + bv.
3. For each $a \in R$a ∈ R and for each pair $v_1, v_2 \in V, a(v_1 + v_2) = av_1 + av_2$.v1, v2 ∈ V , a(v1 + v2) = av1 + av2.
4. For each vector $v \in V$v ∈ V and for each pair $a, b \in R, (ab)v = a(bv)$.a, b ∈ R, (ab)v = a(bv).

Consider a set $V = \{(1, -y) \mid y \in R\} \subseteq R^2$ V = {(1, −y) ∣ y ∈ R} ⊆ R2 with the usual addition and scalar multiplication as in R^2 R2. Answer the following questions (1) and (2).

1 point
Which of the following option(s) is (are) true?

[Hint:Hint: Check the axioms of vector space.]

- ☐ $(1, 2) \in V$(1, 2) ∈ V .
- ☐ $(2, 3) \in V$(2, 3) ∈ V .
- ☐ $V = R^2$ V = R2.
- ☐ $v_1 = (1, -5)$ and $v_2 = (1, 7) \in V \implies v_1 + v_2 \in V$v1 = (1, −5) and v2 = (1, 7) ∈ V ⟹ v1 + v2 ∈ V .
- ☐ $v = (1, 9) \in V, c \in R \implies cv \in V, \forall\, c \in R$v = (1, 9) ∈ V , c ∈ R ⟹ cv ∈ V , ∀ c ∈ R.

☐ For all $v \in V, cv \in V$ For all v ∈ V , cv ∈ V only for c = 1c = 1.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(1, 2) \in V$(1, 2) ∈ V .
For all $v \in V, cv \in V$ For all v ∈ V , cv ∈ V only for c = 1c = 1.

1 point
Which of the following option(s) is (are) true?

☐ Zero element does not exist in VV .
☐ Zero element of VV is (0,0).
☐ Inverse element of (1, 5) is (-1, -5).
☐ Inverse element of (1, 5) is (1, -5).
☐ Inverse element of (1, 5) does not exist.
☐ Inverse element of any element of VV does not exist.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Zero element does not exist in VV .
Inverse element of (1, 5) does not exist.
Inverse element of any element of VV does not exist.

Consider a set $V = \{(x, y, z) \mid x + y + z = 0, x, y, z \in R\} \subseteq R^3$V = {(x, y, z) ∣ x + y + z = 0, x, y, z ∈ R} ⊆ R3 with the usual addition: $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$,(x1, y1, z1) + (x2, y2, z2) = (x1 + x2, y1 + y2, z1 + z2), where $(x_1, y_1, z_1), (x_2, y_2, z_2) \in V$ where (x1, y1, z1), (x2, y2, z2) ∈ V and scalar multiplication: $c(x, y, z) = (cx, cy, cy), \quad c \in R$ and $(x, y, z) \in V$.c(x, y, z) = (cx, cy, cy), c ∈ R and (x, y, z) ∈ V .
Answer the following questions (3) and (4).

1 point
Which of the following option(s) is (are) true?

[Hint:Hint: For a vector $v = (x, y, z)$v = (x, y, z), check whether the condition $x + y + z = 0$ x + y + z = 0 is satisfied or not.]

☐ (1, 2, -3) $\in V$∈ V .
☐ (-1, 2, 1) $\in V$∈ V .
☐ $V = R^3$V = R3.
☐ $(5, -7, 2) \in V$(5, −7, 2) ∈ V .

No, the answer is incorrect.
Score: 0
Accepted Answers:
(1, 2, -3) $\in V$∈ V .

$(5, -7, 2) \in V$(5, −7, 2) ∈ V .

1 point

Inverse element of (-2, -4, 6) with respect to vector addition is

- ○ (2, 4, -6).
- ○ (2, -4, -6).
- ○ (2, 4, 6).
- ○ (-2, 4, -6).

No, the answer is incorrect.

Score: 0

Accepted Answers:

(2, 4, -6).

1 point

Consider a set $V = \{(x, y, z) \mid x + y + z = 0, x, y, z \in R\} \subseteq R^3$ V = {(x, y, z) ∣ x + y + z = 0, x, y, z ∈ R} ⊆ R3 with the usual addition as in R^3 R3 and scalar multiplication is defined as

$$c(x, y, z) = \begin{cases} (0, 0, 0) & c = 0 \\ (\frac{cx}{2}, \frac{y}{c}, cz) & c \neq 0 \end{cases} \quad (x, y, z) \in V,\ c \in R$$c(x, y, z) = {(0, 0, 0)(2cx, cy, cz)

c = 0c = 0 (x, y, z) ∈ V , c ∈ R

Which of the following option(s) is (are) true?

[Hint:Hint: By adding any two elements in VV , or by multiplying a scalar with any element in VV , check whether the resultant element is in VV or not.]

☐ VV is closed under addition i.e., $(v_1, v_2 \in V \implies v_1 + v_2 \in V)$(v1, v2 ∈ V ⟹ v1 + v2 ∈ V).

☐ VV is closed under scalar multiplication i.e., $(c \in R, v \in V \implies cv \in V)$ (c ∈ R, v ∈ V ⟹ cv ∈ V).

☐ VV has zero element with respect to addition.

☐ $1.v = v$1.v = v, where $1 \in R$ and $v \in V$1 ∈ R and v ∈ V .

No, the answer is incorrect.

Score: 0

Accepted Answers:

VV is closed under addition i.e., $(v_1, v_2 \in V \implies v_1 + v_2 \in V)$(v1, v2 ∈ V ⟹ v1 + v2 ∈ V).

VV has zero element with respect to addition.

1 point

Consider usual addition and scalar multiplication on R^2 R2 and the subset $V = \{(1, -y) \mid y \in R\} \subseteq R^2$ V = {(1, −y) ∣ y ∈ R} ⊆ R2.

Which of the following option(s) is (are) true?

[Hint:Hint: Take two arbitrary elements $(1, -y_1)$ and $(1, -y_2)$(1, −y1) and (1, −y2) from VV and check the conditions given in the options.]

☐ For each element $v \in V$v ∈ V and for each pair $a, b \in R$, $(a + b)v = av + bv$a, b ∈ R, (a + b)v = av + bv.

☐ For each $a \in R$a ∈ R and for each pair $v_1, v_2 \in V$, $a(v_1 + v_2) = av_1 + av_2$ v1, v2 ∈ V , a(v1 + v2

) = av1 + av2.

☐ For each element $v \in V$v ∈ V and for each pair $a, b \in R$, $(ab)v = a(bv)$a, b ∈ R, (ab)v = a(bv).

☐ The set VV with the usual addition and scalar multiplication in R^2 R2 is a vector space.

No, the answer is incorrect.

Score: 0

Accepted Answers:

For each element $v \in V$v ∈ V and for each pair $a, b \in R$, $(a + b)v = av + bv$a, b ∈ R, (a + b)v = av + bv.

For each $a \in R$a ∈ R and for each pair $v_1, v_2 \in V$, $a(v_1 + v_2) = av_1 + av_2$ v1, v2 ∈ V , a(v1 + v2) = av1 + av2.

For each element $v \in V$v ∈ V and for each pair $a, b \in R$, $(ab)v = a(bv)$a, b ∈ R, (ab)v = a(bv).

1 point

Consider a set $V = \{(x, y, z) \mid x + y + z = 0, x, y, z \in R\} \subseteq R^3$ V = {(x, y, z) ∣ x + y + z = 0, x, y, z ∈ R} ⊆ R3 with the usual addition as in R^3 R3 and scalar multiplication is defined as

$$c(x, y, z) = \begin{cases} (0, 0, 0) & c = 0 \\ (\frac{cx}{2}, y, cz) & c \neq 0 \end{cases} \quad (x, y, z) \in V,\ c \in R$$c(x, y, z) = {(0, 0, 0)(2cx, cy, cz)

c = 0c = 0 (x, y, z) ∈ V , c ∈ R

Which of the following option(s) is (are) true?

○ For each element $v \in V$v ∈ V and for each pair $a, b \in R$, $(a + b)v = av + bv$a, b ∈ R, (a + b)v = av + bv

○ For each $a \in R$a ∈ R and for each pair $v_1, v_2 \in V$, $a(v_1 + v_2) = av_1 + av_2$ v1, v2 ∈ V , a(v1 + v2) = av1 + av2

○ For each element $v \in V$v ∈ V and for each pair $a, b \in R$, $(ab)v = a(bv)$a, b ∈ R, (ab)v = a(bv)

○ VV is a vector space.

No, the answer is incorrect.

Score: 0

Accepted Answers:

For each $a \in R$a ∈ R and for each pair $v_1, v_2 \in V$, $a(v_1 + v_2) = av_1 + av_2$ v1, v2 ∈ V , a(v1 + v2) = av1 + av2

Key Points: 2Key Points: 2

- Definition of a subspace: A subset WW of a vector space VV is a subspace of VV if it is a vector space with respect to the same operations defined on VV .
- Criteria to check a subset WW of a vector space VV is a subspace:

1. Zero vector should be in WW i.e., $0 \in W$0 ∈ W.
2. If two vectors v_1 and $v_2 \in W$v1 and v2 ∈ W, then $v_1 + v_2 \in W$v1 + v2 ∈ W.
3. Let $c \in R$c ∈ R and if $v \in W$v ∈ W, then $cv \in W$cv ∈ W

1 point

Let $W = \{(x, y) \mid x + 3y = 1, x, y \in R\} \subseteq R^2$W = {(x, y) | x + 3y = 1, x, y ∈ R} ⊆ R2. Which of the following options are true?

- [ ] Zero element exists in WW.
- [ ] If v_1 and $v_2 \in W$v1 and v2 ∈ W, then $v_1 + v_2 \notin W$v1 + v2 ∉/ W.
- [ ] Let $c \in R$c ∈ R and if $v \in W$v ∈ W, then $cv \in W$cv ∈ W
- [ ] WW is not a subspace of R^2 R2.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If v_1 and $v_2 \in W$v1 and v2 ∈ W, then $v_1 + v_2 \notin W$v1 + v2 ∉/ W.
WW is not a subspace of R^2 R2.

1 point

Let $W = \{A \mid A \in M_{2\times 2}(R)$ and A is a diagonal matrix}W = {A | A ∈ M2×2 (R) and A is a diagonal matrix} be a subset of a vector space $M_{2\times 2}(R)$M2×2(R). Which of the following option(s) is (are) true?

- [ ] Zero vector does not exist in WW.
- [ ] If v_1 and $v_2 \in W$v1 and v2 ∈ W, then $v_1 + v_2 \in W$v1 + v2 ∈ W.
- [ ] Let $c \in R$c ∈ R and if $v \in W$v ∈ W, then $cv \in W$cv ∈ W
- [ ] WW is a subspace of $M_{2\times 2}(R)$M2×2(R).

No, the answer is incorrect.
Score: 0
Accepted Answers:
If v_1 and $v_2 \in W$v1 and v2 ∈ W, then $v_1 + v_2 \in W$v1 + v2 ∈ W.
Let $c \in R$c ∈ R and if $v \in W$v ∈ W, then $cv \in W$cv ∈ W
WW is a subspace of $M_{2\times 2}(R)$M2×2(R).

Key Points: 3Key Points: 3

- Linear combination: Let VV be a vector space and $v_1, v_2, ...v_k \in V$v1, v2, ...vk ∈ V . The linear combination of $v_1, v_2, ...v_k$ v1, v2, ...vk with coefficients $a_1, a_2, ...a_k \in R$a1, a2, ...ak ∈ R is the vector $v = \sum_{i=1}^{k} a_i v_i$ v = ∑i=1k aivi.
- A vector $v \in V$v ∈ V is said to be a linear combination of $v_1, v_2, ...v_k \in V$v1, v2, ...vk ∈ V if there exist $a_1, a_2, ...a_k \in R$a1, a2, ...ak ∈ R so that $v = \sum_{i=1}^{k} a_i v_i$ v = ∑i=1k aivi

1 point

Which of the following vectors are linear combinations of (2, 1) and (6, 3) with usual addition and scalar multiplication in R^2 R2?

- [ ] (2, 1) + 5(6, 3)
- [ ] (2, 1)
- [ ] (6, 3)
- [ ] (3, 6)

No, the answer is incorrect.
Score: 0
Accepted Answers:
(2, 1) + 5(6, 3)
(2, 1)
(6, 3)
1 point
Let VV be a vector space and v_1, v_2, v_3 and $v_4 \in V$v1, v2, v3 and v4 ∈ V . If v_1v1 is a linear combination of $v_i, i = 2, 3, 4$ $i.e.,\ v_1 = av_2 + bv_3 + cv_4$vi, i = 2, 3, 4 i.e., v1 = av2 + bv3 + cv4, then which of the following options are true?

☐ If $a \neq 0$a = 0, then v_2v2 is a linear combination of $v_i, i = 1, 3, 4$vi, i = 1, 3, 4.

☐ If $a = 0$a = 0 and $b \neq 0$b = 0, then v_3v3 is a linear combination of $v_i, i = 1, 4$vi, i = 1, 4.

☐ v_2v2 is a linear combination of v_2v2.

☐ None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If $a \neq 0$a = 0, then v_2v2 is a linear combination of $v_i, i = 1, 3, 4$vi, i = 1, 3, 4.
If $a = 0$a = 0 and $b \neq 0$b = 0, then v_3v3 is a linear combination of $v_i, i = 1, 4$vi, i = 1, 4.
v_2v2 is a linear combination of v_2v2.